affirmoTM tutorial 4: running the model

What tm_run() does, and the steps of a run one at a time

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This tutorial starts from a fresh analysis folder with the example input files (see tutorial 1):

```
library(affirmoTM)
```

```
tm_example_inputs()
```

1 What `tm_run()` does

`tm_run()` runs the whole model with the files in `inputs/`:

1. reads and checks every input file and setting; any problem stops it before anything is run or written;
2. lists the death states it found in the matrices;
3. runs the model `n_runs` times, run r with seed `seed + r - 1`;
4. averages the runs, compares the arms, and writes the results to `outputs/`.

While it runs, it reports the death states and its progress (here with 20 runs):

```
settings <- tm_read_settings()
settings$n_runs <- 20
results <- tm_run(settings)
```

```
Death states: CurrentDeath, AB_Death, AHS_Death, AIS_Death, AHS_AB_Death, AIS_AB_Death.

Run 10 of 20 done.

Run 20 of 20 done.
```

Check that no living state is among the death states.

1.1 Run time

The run time grows with the number of runs and the number of patients:

```
system.time(tm_run(settings))
```

```
   user  system elapsed
 26.444   0.249   6.381
```

1.2 The same seed, the same results

```
again <- tm_run(settings)
identical(results$by_run, again$by_run)
```

```
[1] TRUE
```

1.3 Outputs

Every call writes, and overwrites, the files in `outputs/`:

```
list.files("outputs")
```

```
[1] "ce_plane.png"          "comparison_by_run.csv" "comparison.csv"
[4] "summary_by_run.csv"    "summary.csv"
```

Tutorial 5 explains the results.

2 The steps of a run, one at a time

`tm_run()` calls the functions below, once per run. They can also be used one at a time, for example to look at one run in detail. Each takes the `settings` list. Tutorial 6 follows a run step by step in more detail.

Read the inputs. Each reader checks its file:

```
matrices <- tm_read_matrices(settings)
cohort <- tm_read_cohort(settings, matrices)
entry_states <- tm_read_entry_states(settings, matrices)
costs <- tm_read_costs(settings, matrices)
utilities <- tm_read_utilities(settings, matrices)
intervention <- tm_read_intervention_cost(settings)
death <- matrices$usual_care$death
```

Entry states. `tm_assign_entry_states()` draws each patient's state in their entry year. Both arms start from these states:

```
patients <- tm_assign_entry_states(cohort, entry_states, settings, seed = settings$seed)
patients[, .N, keyby = entry_state]
```

```
Key: <entry_state>
     entry_state     N
          <char> <int>
 1:           AB    17
 2:          AHS    16
 3:       AHS_AB     1
 4:          AIS    64
 5: CurrentDeath   420
 6:      NoEvent  4318
 7:           PB    43
 8:       PB_AIS     2
 9:          PHS    28
10:          PIS    89
11:        PS_AB     2
```

Simulation. `tm_simulate()` moves each patient through both arms, one year at a time, until they die or the horizon ends. One row per patient, arm and year:

```
sim <- tm_simulate(patients, matrices, settings, seed = settings$seed)
sim[id == 1]
```

```
       id        arm cycle   age agegrp    sex       mm_cluster   state
    <int>     <char> <int> <int> <char> <char>           <char>  <char>
 1:     1 usual_care     0    65  65-69    men Neuropsychiatric NoEvent
 2:     1 usual_care     1    66  65-69    men Neuropsychiatric NoEvent
 3:     1 usual_care     2    67  65-69    men Neuropsychiatric NoEvent
 4:     1 usual_care     3    68  65-69    men Neuropsychiatric NoEvent
 5:     1 usual_care     4    69  65-69    men Neuropsychiatric NoEvent
 6:     1 usual_care     5    70  70-74    men Neuropsychiatric NoEvent
 7:     1 usual_care     6    71  70-74    men Neuropsychiatric NoEvent
 8:     1 usual_care     7    72  70-74    men Neuropsychiatric NoEvent
 9:     1 usual_care     8    73  70-74    men Neuropsychiatric NoEvent
10:     1 usual_care     9    74  70-74    men Neuropsychiatric NoEvent
11:     1       iabc     0    65  65-69    men Neuropsychiatric NoEvent
12:     1       iabc     1    66  65-69    men Neuropsychiatric NoEvent
13:     1       iabc     2    67  65-69    men Neuropsychiatric NoEvent
14:     1       iabc     3    68  65-69    men Neuropsychiatric NoEvent
15:     1       iabc     4    69  65-69    men Neuropsychiatric NoEvent
16:     1       iabc     5    70  70-74    men Neuropsychiatric NoEvent
17:     1       iabc     6    71  70-74    men Neuropsychiatric NoEvent
18:     1       iabc     7    72  70-74    men Neuropsychiatric NoEvent
19:     1       iabc     8    73  70-74    men Neuropsychiatric NoEvent
20:     1       iabc     9    74  70-74    men Neuropsychiatric NoEvent
       id        arm cycle   age agegrp    sex       mm_cluster   state
    <int>     <char> <int> <int> <char> <char>           <char>  <char>
```

Costs and QALYs. `tm_add_outcomes()` gives each year its cost, intervention cost, QALY and discount factor:

```
out <- tm_add_outcomes(sim, costs, utilities, intervention, death, settings)
out[id == 1, .(arm, cycle, state, cost, intervention_cost, qaly, discount)]
```

```
         arm cycle   state  cost intervention_cost  qaly  discount
      <char> <int>  <char> <num>             <num> <num>     <num>
 1:     iabc     0 NoEvent  2000                 0  0.75 1.0000000
 2:     iabc     1 NoEvent  2050                50  0.75 0.9708738
 3:     iabc     2 NoEvent  2000                 0  0.75 0.9425959
 4:     iabc     3 NoEvent  2000                 0  0.75 0.9151417
 5:     iabc     4 NoEvent  2000                 0  0.75 0.8884870
 6:     iabc     5 NoEvent  2000                 0  0.75 0.8626088
 7:     iabc     6 NoEvent  2000                 0  0.75 0.8374843
 8:     iabc     7 NoEvent  2000                 0  0.75 0.8130915
 9:     iabc     8 NoEvent  2000                 0  0.75 0.7894092
```

```
10:       iabc     9 NoEvent  2000                0  0.75 0.7664167
11: usual_care     0 NoEvent  2000                0  0.75 1.0000000
12: usual_care     1 NoEvent  2000                0  0.75 0.9708738
13: usual_care     2 NoEvent  2000                0  0.75 0.9425959
14: usual_care     3 NoEvent  2000                0  0.75 0.9151417
15: usual_care     4 NoEvent  2000                0  0.75 0.8884870
16: usual_care     5 NoEvent  2000                0  0.75 0.8626088
17: usual_care     6 NoEvent  2000                0  0.75 0.8374843
18: usual_care     7 NoEvent  2000                0  0.75 0.8130915
19: usual_care     8 NoEvent  2000                0  0.75 0.7894092
20: usual_care     9 NoEvent  2000                0  0.75 0.7664167
          arm cycle   state  cost intervention_cost  qaly  discount
       <char> <int>  <char> <num>             <num> <num>     <num>
```

Summary. `tm_summarise()` adds up each arm:

```
one_run <- tm_summarise(out, death, settings)
one_run[, .(arm, patients, cost_per_patient, qaly_per_patient, deaths)]
```

```
         arm patients cost_per_patient qaly_per_patient deaths
      <char>    <int>            <num>            <num>  <int>
1:       iabc     5000         13123.60         3.593847   1845
2: usual_care     5000         13219.94         3.589423   1846
```

Comparison. `tm_compare()` takes iABC minus usual care:

```
tm_compare(one_run[, `:=`(run = 1L, seed = settings$seed)])$overall
```

```
   delta_cost delta_intervention_cost  delta_qaly      icer quadrant
        <num>                   <num>       <num>     <num>   <char>
1:  -96.34138                37.44662 0.004423595 -21778.98 dominant
   delta_ischaemic_stroke delta_haemorrhagic_stroke delta_major_bleed delta_deaths
                    <num>                     <num>             <num>        <num>
1:                    -33                       -10                -1           -1
```

This is exactly the first run of `tm_run()`, which uses the same seed:

```
results$by_run[run == 1, .(arm, patients, cost_per_patient, qaly_per_patient, deaths)]
```

```
         arm patients cost_per_patient qaly_per_patient deaths
      <char>    <int>            <num>            <num>  <int>
1:       iabc     5000         13123.60         3.593847   1845
2: usual_care     5000         13219.94         3.589423   1846
```